BAN: A Boundary-Aware Network Based on KAN for Robust Hyperspectral Image Classification Against Adversarial Attacks

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***Abstract*—Deep neural networks (DNNs) have achieved significant advancements in hyperspectral image (HSI) classification, enabling critical applications in environmental monitoring, medical imaging, and geological analysis. However, their vulnerability to adversarial attacks, particularly in the boundary regions between classes, remains a critical challenge. Existing defense methods often struggle to address these issues due to insufficient adaptability to irregular boundary patterns and larger perturbations in boundary regions. To bridge this gap, we propose a boundary-aware network (BAN), a novel adversarial defense network framework that integrates a multiscale deformable convolution (MSDC) module with a Kolmogorov–Arnold network (KAN). The MSDC module dynamically adjusts receptive fields across scales to capture discriminative spatial-spectral features, while the KAN architecture leverages its cubic spline-based smooth nonlinearity to suppress gradient-driven adversarial perturbations. By leveraging these components, BAN not only mitigates boundary-specific vulnerabilities but also enhances global robustness against diverse adversarial threats. Experimental results on three benchmark HSI datasets demonstrate that BAN outperforms state-of-the-art methods, maintaining high accuracy and robustness under various adversarial attack scenarios.**

***Index Terms*—Adversarial attack, adversarial defense, deformable convolution, hyperspectral image (HSI) classification, Kolmogorov–Arnold network (KAN).**

I. Introduction

With the rapid advancements of deep neural networks (DNNs), hyperspectral image (HSI) analysis has made remarkable progress across various fields, including geological exploration [1], environmental monitoring [2], [3], and medical image processing [4], [5]. These advancements have enabled a wide range of applications, such as mineral identification [6], land cover classification [7], and lesion detection [8], [9], significantly expanding the potential of HSI technology. Despite the growing importance of DNNs, they remain highly susceptible to adversarial examples, which are generated by adding imperceptible perturbations to inputs. While these perturbations are imperceptible to humans and do not affect human recognition, they can cause DNNs to produce incorrect predictions with high confidence. For instance, adversarial attacks on traffic signs can lead to severe accidents [10], [11], [12], and those targeting facial recognition systems might compromise personal privacy [13], [14]. These vulnerabilities of DNNs highlight the urgent need for robust defense mechanisms against adversarial threats.

The susceptibility of DNNs to adversarial attacks in the remote sensing field was first highlighted by Czaja et al. [15], with subsequent studies [16], [17], [18] confirming their prevalence in HSI classification tasks. To address this issue, several defense strategies have been proposed. Xu et al. [19] introduced the self-attention context network (SACNet) to capture global spatial relationships and distribute perturbation risks across the entire HSI. Tu et al. [20] improved context modeling between intra-class and inter-class features through a class-context-aware learning module, reducing the impact of perturbations. Tang et al. [21] proposed a weighted fusion method that combines spectral transformer and spatial self-attention (WFSS) to capture the relationships between various spectral bands. Xu et al. [22] developed the spatial–spectral self-attention learning network (S^3ANet), specifically designed to defend against adversarial attacks in HSI classification while addressing the scale-specific object classification needs. In our previous research, we presented a groundbreaking antiperturbation network referred to as APNet [23], which significantly enhances the robustness of feature representations against adversarial attacks by fusing denoised pixel-level features with coarse-grained superpixel-level features.

Although these methods have made significant progress in defending against adversarial attacks on HSI classification networks, few have fully explored the impact of these attacks and provided corresponding strategies. As illustrated in Fig. 1, the impact of such attacks on HSI classification is predominantly concentrated in the boundary regions between categories. This phenomenon can be attributed to two main factors: first, most existing classification networks are constrained by limited and fixed receptive fields, preventing them from adequately addressing the complex and irregular boundary areas. Second,

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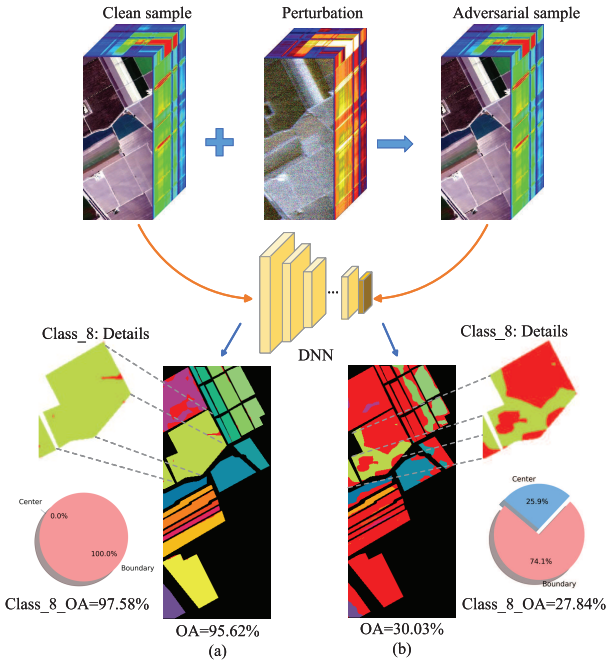


Fig. 1. Schematic illustration of adversarial attacks on DNN-based HSI classification. (a) and (b) OA of a DNN-based classifier on clean and adversarial samples, respectively. The zoomed-in area details the classification results for Class 8, wherein a pie chart further elucidates the proportion of misclassified pixels in the central and boundary regions within Class 8.

gradient-driven attacks tend to generate larger perturbations in these boundary regions due to significant feature changes, making it challenging for networks to effectively mitigate or filter out perturbations originating from the boundaries.

To address the challenges, we propose a boundary-aware network (BAN), a novel adversarial defense network that leverages multiscale deformable convolution (MSDC) module with learnable pixel offsets to capture spatial–spectral features at different scales and reduce the impact of gradient-driven perturbations through cubic spline-based Kolmogorov–Arnold network (KAN) [24], [25], [26], [27] architecture. The key innovations of BAN include utilizing the intricate nonlinear modeling ability and high smoothness of the KAN structure and designing an MSDC module to capture spatial–spectral features with key structural information. Specifically, BAN first employs the MSDC to extract multiscale spatial–spectral features. which are then fed into the KAN network to capture intricate nonlinear interactions while maintaining smooth gradients. By effectively extracting robust representation and suppressing adversarial perturbations, BAN achieves a comprehensive and effective adversarial defense. The main contributions of this article are as follows.

1) To the best of our knowledge, this is the first attempt to reveal the phenomenon of high attack success rates in the boundary regions of HSI classification, and we propose BAN, a robust adversarial defense network that leverages MSDC module and KAN architecture to address the susceptibility of DNNs to adversarial attacks in HSI classification, especially in the boundary regions.
2) We introduce the MSDC module, which combines deformable and dilated convolutions across multiple scales to effectively extract spatial–spectral features with key structural information, particularly enhancing the capture of complex patterns within boundary regions.
3) To further improve adversarial robustness, also for the first time, we explore the intricate nonlinear modeling ability and high smoothness of KAN to adaptively learn complex spatial relationships and reduce the impact from gradient-driven perturbations, thereby achieving a robust representation and effective adversarial defense.
4) We conduct extensive experiments on three widely used HSI datasets to evaluate the performance of BAN under various types and intensities of adversarial attacks. The results demonstrate that BAN outperforms state-of-the-art methods, maintaining high accuracy and robustness even under high-intensity attacks.

The remainder of this article is organized as follows. Section II reviews adversarial attacks, defense methods, robust and non-robust features. Section III elaborates on the proposed BAN in detail. Section IV presents the performance, comparisons, and analyses of this study on three benchmark datasets. Finally, Section V concludes this article.

II. Related Works

A. Adversarial Attack

1) Fast Gradient Sign Method: Goodfellow et al. [28] first introduced the fast gradient sign method (FGSM) and discussed the impact of adversarial examples on deep learning models. FGSM generates adversarial examples quickly by calculating the gradient of the loss function with respect to the input data, using the sign function to obtain the most significant direction of the gradient, and making small adjustments based on that direction. Specifically, FGSM uses the following formula to generate adversarial examples:

$$x^{\text{adv}} = \text{clip}\left(x + \epsilon \cdot \text{sign}\left(\nabla_x J\left(\theta, x, y\right)\right)\right) \tag{1}$$

where θ represents the model parameters, x represents the original input data, y is the true label, and $\nabla_x J()$ denotes the gradient of the loss function J with respect to the input x, indicating how to change x to increase the loss. ε represents the magnitude of the perturbation, while clip$(\cdot)$ ensures that the pixels of the adversarial example remain within a reasonable range after the perturbation is added.

2) Project Gradient Descent: Project gradient descent (PGD) is an extension of FGSM, which enhances the effectiveness of the attack by performing multiple iterations with smaller perturbations [29]. It projects the adversarial samples learned during each iteration back into the specified range, thereby increasing the effectiveness of the attack. The adversarial samples generated in each iteration are represented as follows:

$$x_{t+1}^{\text{adv}} = \text{clip}\left(x_t^{\text{adv}} + \epsilon \cdot \text{sign}\left(\nabla_x J\left(\theta, x_t, y\right)\right)\right) \tag{2}$$

where x_{t+1}^{adv} denotes the adversarial sample generated at iteration t. When $t = 0$, x_0 is a clean sample. The definitions of other parameters are the same as those in the aforementioned FGSM method.

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3) Carlini and Wagner's Attack: Carlini and Wagner's attack (C&W) aims to find the minimal perturbation between adversarial samples and clean samples by optimizing a specific objective function [30], thereby generating adversarial samples with minimal perturbation. The optimization process is outlined as follows:

$$\begin{aligned} &\text{minimize } D(x+\delta, x) \\ &\text{such that } C(x+\delta) = t \\ &x+\delta \in [0,1]^n \end{aligned} \tag{3}$$

where D is a function that measures the distance between the original image and the adversarial sample. Our goal is to find the perturbation δ that minimizes $D(x^{\text{adv}}, x)$, while satisfying the two constraints $C(x+\delta) = t$ and $x+\delta \in [0,1]^n$. C is the classification model, and t represents the specific target class. Additionally, the perturbation $\delta \in [0,1]^n$ is constrained to control the range of perturbation. This method effectively balances the accuracy of generated adversarial samples with the minimization of perturbation.

B. Adversarial Defense

1) Adversarial Sample Detection: Adversarial sample detection aims to identify and reject adversarial inputs at inference time, typically employing detection mechanisms or anomaly detection techniques to distinguish between normal and adversarial inputs [31]. Xu et al. [32] proposed a feature compression method that reduces the attack search space by merging samples with different feature vectors. By comparing the predictions of DNN models on the original and compressed inputs, feature compression can effectively detect adversarial samples with high accuracy and low false positive rates.

2) Adversarial Training: Adversarial training enhances the robustness of a model by introducing adversarial samples during the training phase. Cheng et al. [33] proposed perturbation-seeking generative adversarial networks (PSGANs), a method that trains classifiers by incorporating samples generated during the image reconstruction process. Subsequently, Su et al. [34] proposed an improved adversarial training method that combines reconstruction assistance and distance optimization, aiming to eliminate perturbations in adversarial samples and improve training performance. It is noteworthy that, while adversarial training can enhance the adversarial robustness of the network, it comes at the cost of increased computational burden and may lead to a decrease in recognition accuracy on clean samples.

3) Adversarial Defense Networks: To overcome the limitations of these two strategies, Qi et al. [35] proposed a masked spatial–spectral autoencoder (MSSA) based on self-supervised learning theory, which significantly enhances the robustness of HSI analysis systems by integrating mask sequence attention learning with learnable graph structures. Liu et al. [36] introduced a denoising diffusion defense recovery network (D3R-Net), which transforms adversarial samples into a noisy state using an efficient denoising diffusion probabilistic model. The system then recovers semantic information through a series of iterative denoising steps, thereby significantly improving the network's defense capability against unknown attacks.

C. Robust and Non-Robust Features

We define a feature as a set of functions that map the input space x to real numbers, and thus the feature space is denoted as $F = \{x \to \mathbb{R}\}$. To ensure scale invariance, we assume that all features in the set F are subjected to translation and scaling such that their mean is 0 and their variance is 1, as expressed by the following formula:

$$\begin{aligned} \mathbb{E}_{(x,y)\sim D}\left[f(x)\right] &= 0 \\ \mathbb{E}_{(x,y)\sim D}\left[f(x)^2\right] &= 1. \end{aligned} \tag{4}$$

For a given data distribution D, we refer to a feature f as useful if it is positively correlated with the true label in expectation, that is, if it satisfies

$$\mathbb{E}_{(x,y)\sim D}\left[y \cdot f(x)\right] \geq \rho \tag{5}$$

where x represents the input sample, y denotes the true label, $f(x)$ represents the feature value, and $\rho(\rho > 0)$ is the threshold that quantifies the strength of the correlation between the feature $f(x)$ and the label y.

Assume we have a ρ-useful feature $f(\rho D(f) > 0)$. We define f as a robustly useful feature if f remains useful under adversarial perturbation ε. Formally,

$$\mathbb{E}_{(x,y)\sim D}\left[\inf_{\delta\in\Delta(x)} y \cdot f(x+\delta)\right] \geq \gamma. \tag{6}$$

This means that, among all adversarial perturbations ε, the expected minimum value of the product between the feature f and the label y remains greater than or equal to $\gamma(\gamma > 0)$, and f is considered a useful and robust feature.

A useful but non-robust feature refers to a ρ-useful feature that is not γ-robust. These features aid in classification under normal conditions but hinder accuracy in adversarial situations because their correlation with the label can be flipped.

III. Proposed Method

A. Overview

Our research indicates that in the context of HSI classification, even minor adversarial perturbations can lead to substantial misclassification issues, especially within the transitional zones at the boundaries between classes. Current defense methodologies fall short in adaptively addressing the complexity and irregularity of boundary regions due to being limited by finite and fixed receptive fields. To address this challenge, we propose BAN, an innovative adversarial defense network. BAN utilizes MSDC with learnable offsets to capture spatial–spectral features at various scales and reduces the impact of gradient-driven perturbations through a KAN structure based on cubic splines. We outline the proposed BAN framework in Fig. 2. Specifically, the MSDC extracts multiscale spatial–spectral features using deformable convolutions and dilated convolutions and then feeds these features into the KAN structure to capture complex nonlinear interactions and maintain smooth gradients. After the KAN layer, these features are passed through an efficient depth-wise convolutional (DwConv) layer.

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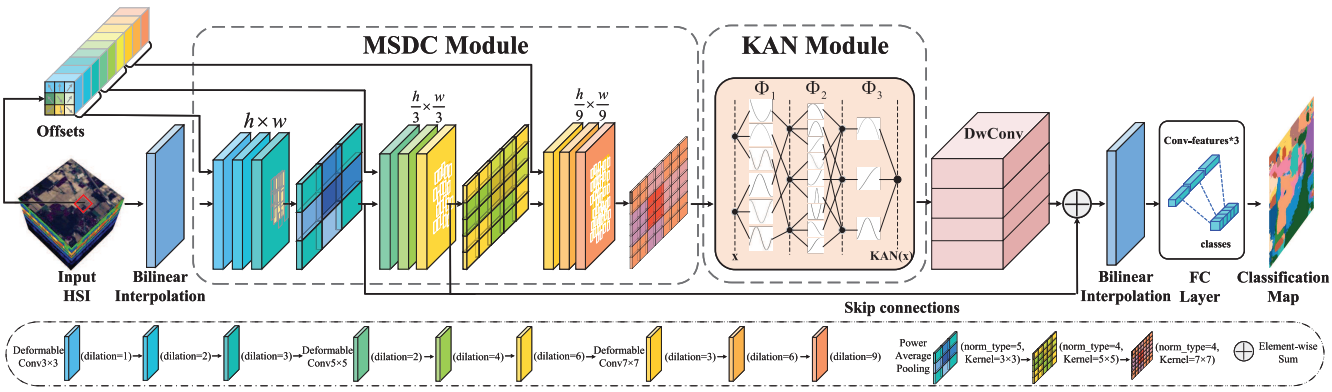


Fig. 2. Overview of the proposed BAN architecture. The MSDC module employs multibranch deformable convolutions to enhance its ability to capture edge features with learnable offsets. Processed features then pass through a hierarchical KAN module that performs nonlinear modeling through cubic spline-based transformations, followed by DwConv layers for cross-channel fusion. The classification map is ultimately produced via bilinear interpolation and a fully connected layer.

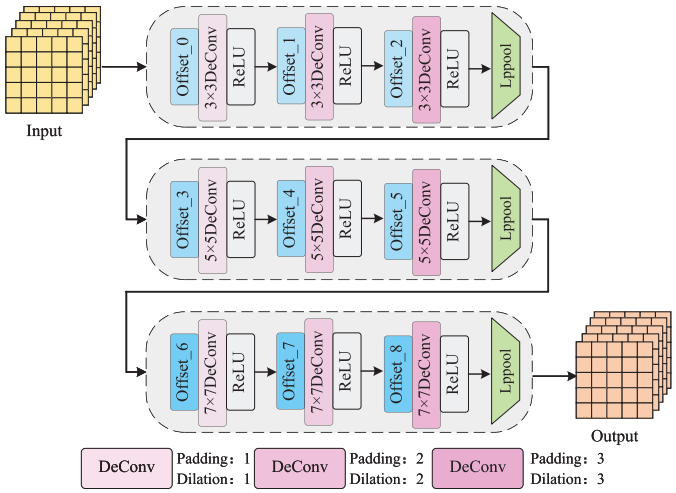


Fig. 3. Structure of the proposed MSDC module.

B. MSDC Module for Boundary-Aware Representation

Conventional convolutional neural networks (CNNs) are inherently limited by their restricted receptive fields and fixed scale processing capabilities, which hinder their ability to effectively handle boundary regions. These limitations make it particularly challenging for CNNs to mitigate or filter out perturbations near class boundaries, especially when faced with substantial disturbances.

To address this challenge, we propose an MSDC module, as illustrated in Fig. 3, extending the deformable convolution network (DCN) proposed by Dai et al. [37]. Specifically, the MSDC architecture introduces three key improvements: in contrast to the single convolutional layer rigid offset prediction of DCN, the proposed MSDC establishes a dedicated offset generation network that adaptively learns the deformation domain under hierarchical feature conditions. By limiting the offsets based on tanh activation within [−1, 1], the MSDC addresses the unstable gradient propagation of the DCN while maintaining numerical stability. Critically, the incorporation of the task-driven L_2 regularization adaptively controls the parametric smoothness of deformation fields.

DCN extends traditional convolution by introducing learnable offsets for each pixel. In contrast to the single-kernel structure of DCN, the offsets in MSDC are generated through the application of a multibranch convolutional structure on the input feature maps. The learnable offset $\Delta p_n^{(k)}$ for the kth group can be expressed as

$$\Delta p_n^{(k)} = \tanh\left(\mathrm{Conv}_{\mathrm{offset}}^{(k)}(X)\right) \tag{7}$$

here, $\mathrm{Conv}_{\mathrm{offset}}^{(k)}(\cdot)$ denotes the kth offset generator and the $\tanh(\cdot)$ activation function constrains the offsets within the range $[-1, 1]$. To further enhance the adaptive regulation of the offsets, an L_2 regularization loss is introduced

$$L_2 = \lambda \sum_{k=1}^{K} \sum_{n} \left\| \Delta p_n^{(k)} \right\|_2^2 \tag{8}$$

where λ is the regularization coefficient that controls the strength of the deformation and constraint intensity. The deformable convolution calculation for MSDC can be represented as

$$y(p_0) = \sum_{p_n \in R_k} W(p_n) X\left(p_0 + d \cdot p_n + \Delta p_n^{(k)}\right). \tag{9}$$

Here, p_0 represents the center of the convolutional window, R_k denotes the set of predefined relative positions within the window, p_n is an element of R_k, and d is the dilation rate of the convolution kernel. The weight of the convolution kernel at position p_n is represented by $W(p_n)$.

For a clearer understanding, the MSDC is composed of three progressively larger deformable convolution blocks with kernel sizes of 3×3, 5×5, and 7×7. Each block $f_D^{k\times k}(\cdot)$ follows the structure {Deformable Conv $k \times k$, Dilation, ReLU}, where D denotes the dilation rate and ReLU is the rectified linear unit activation function. The detailed procedure is as follows:

$$\begin{aligned} \mathrm{block}_1 &= P_{\mathrm{avg}}\left(f_D^{3\times 3}(X_{\mathrm{adv}})\right) \\ \mathrm{block}_2 &= P_{\mathrm{avg}}\left(f_D^{5\times 5}(\mathrm{block}_1)\right) \\ X_{\mathrm{MSDC}} &= P_{\mathrm{avg}}\left(f_D^{7\times 7}(\mathrm{block}_2)\right). \end{aligned} \tag{10}$$

When integrated with power average pooling $P_{\mathrm{avg}}(\cdot)$, the MSDC further refines the sensitivity to input features by

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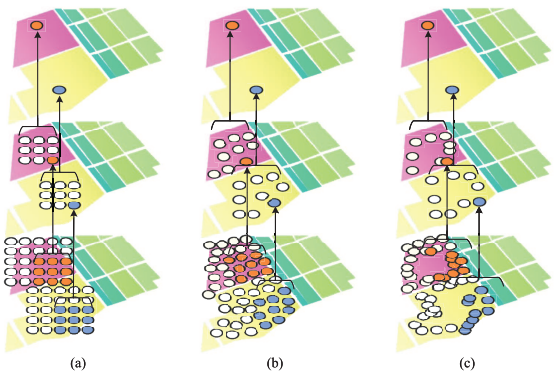


Fig. 4. Illustration of the three convolutional sampling approaches. (a) Fixed geometric sampling in standard convolution. (b) Dynamic offset sampling in deformable convolution. (c) Adaptive multilevel sampling in MSDC.

leveraging customizable exponential parameters. By adjusting tunable parameters, it further refines the sensitivity to input features, enabling adaptive adjustment of the feature response distribution. This retains more original feature information and enhances performance in terms of precision and robustness in boundary regions.

To further illustrate the differences in deformable offset learning between our proposed MSDC and the deformable convolution, Fig. 4 presents three convolutional sampling approaches.

1) Standard convolution employs fixed sampling grids, which limit its ability to handle objects with complex geometric transformations.
2) Deformable convolution improves geometric deformation modeling by introducing learnable offsets.
3) MSDC further enhances edge feature extraction by constraining the effective range of offsets, thereby improving the accuracy and robustness of feature capture in boundary regions while preserving the original feature representation.

C. KAN Module

Traditional neural networks rely heavily on the specific structure of linear weight matrices and employ nonlinear activation functions to capture complex patterns, such as ReLU and GELU. The ReLU function exhibits discontinuous first-order derivatives at zero, leading to gradient mutation regions in the input space. While GELU is smooth, its high-order derivatives may exhibit significant variability. In contrast, the second-order continuous differentiability of cubic B-spline functions enables smoother gradient propagation. This property significantly reduces gradient variance during backpropagation and smooths the transmission of input perturbations through activation functions.

Let $c_{-1}, c_0, \ldots, c_{m+1}$ be a sequence of control points and $t_i = ih$ $(i = 0, \ldots, m)$ be an isometric sequence of nodes. The cubic B-spline function $S(x)$ provided by Cox–de Boor in [38] is composed of piecewise polynomials of degree 3 (which means order 4). Specifically,

$$S(x) = \sum_{i=-1}^{m+1} c_i \varphi_{i,3}(x) = \sum_{i=-1}^{m+1} c_i B_3\left(\frac{x - t_i}{h}\right), \quad x \in [t_0, t_m]. \tag{11}$$

Here, $\varphi_{i,3}(x)$ is the basis function defined on the ith node, which is equal to the central parent function $B_3(u)$, and $\varphi_{i,3}(x) = B_3(u; i)$, where $u(x; i) := (x - x_i)/h \in [-2, 2]$. The recursive relationship for the parent functions $B_k(u)$ is defined as follows:

$$kB_k(u) = \left(\frac{k+1}{2} - u\right) B_{k-1}\left(u - \frac{1}{2}\right) + \left(\frac{k+1}{2} + u\right) B_{k-1}\left(u + \frac{1}{2}\right), \quad k = 1, 2, 3, \ldots \tag{12}$$

The parent function of order 1 (which corresponds to degree 0) is given by

$$B_0(u) = \begin{cases} 1, & -1/2 \le u \le 1/2 \\ 0, & \text{otherwise.} \end{cases} \tag{13}$$

Based on the recursive formula of B-spline parent functions, the cubic B-spline sub-interval function in the ith unit $[t_i, t_{i+1}]$ can be defined by

$$S_i(u_{i-1}, u_i, u_{i+1}, u_{i+2}) = \begin{cases} \frac{1}{6}\begin{bmatrix} \left(8 - 12u_{i-1} + 6u_{i-1}^2 - u_{i-1}^3\right) c_{i-1} \\ + \left(4 - 6u_i^2 + 3u_i^3\right) c_i \\ + \left(4 - 6u_{i+1}^2 - 3u_{i+1}^3\right) c_{i+1} \\ + \left(8 + 12u_{i+2} + 6u_{i+2}^2 + u_{i+2}^3\right) c_{i+2} \end{bmatrix}, & u \in [-2, 2] \\ 0, & \text{otherwise} \end{cases} \tag{14}$$

where $u^{(i)} := u^{(i)}(x) \in [-2, 2]$ and the cubic B-spline function $S(x)$ is given by

$$S(x) = \begin{cases} S_i\left(u^{(i-1)}, u^{(i)}, u^{(i+1)}, u^{(i+2)}\right), & x \in [t_i, t_{i+1}] \\ 0, & \text{otherwise.} \end{cases} \tag{15}$$

Using the derivative formula of the polynomials, the first-order and second-order derivatives of the B-spline basis function are derived as follows:

$$S_i'(u_{i-1}, u_i, u_{i+1}, u_{i+2}) = \begin{cases} \frac{1}{2}\begin{bmatrix} \left(-4 + 4u_{i-1} - u_{i-1}^2\right) c_{i-1} \\ + \left(-4u_i + 3u_i^2\right) c_i \\ + \left(-4u_{i+1} - 3u_{i+1}^2\right) c_{i+1} \\ + \left(4 + 4u_{i+2} + u_{i+2}^2\right) c_{i+2} \end{bmatrix}, & u \in [-2, 2] \\ 0, & \text{otherwise} \end{cases} \tag{16}$$

$$S_i''(u_{i-1}, u_i, u_{i+1}, u_{i+2}) = \begin{cases} \begin{bmatrix} (2 - u_{i-1}) c_{i-1} \\ + (-2 + 3u_i) c_i \\ + \left(-2 - 3u_{i+1}\right) c_{i+1} \\ + \left(2 + u_{i+2}\right) c_{i+2} \end{bmatrix}, & u \in [-2, 2] \\ 0, & \text{otherwise.} \end{cases} \tag{17}$$

Therefore, it is evident that both the first-order and second-order derivatives of a cubic B-spline function are continuously differentiable. The variance of the loss function

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of backward gradient propagation based on B-spline functions, $\varepsilon_{B-\text{spline}}(\nabla L)$, is expressed as

$$\varepsilon_{B-\text{spline}}(\nabla L) \propto O\left[\left(S''(x)\cdot\Delta x\right)^2\right] - \left[O\left(S''(x)\cdot\Delta x\right)\right]^2. \quad (18)$$

Here, Δx is the input perturbation and $O(\cdot)$ is the calculation of the truncation error. Since $S''(x)$ is a continuous, bounded piecewise linear function, its gradient change satisfies constant boundlessness, and the mean of the perturbation is zero. Then the errors of the B-spline loss function can be updated as

$$\varepsilon_{B-\text{spline}}(\nabla L) \propto O\left[\left(S''(x)\cdot\Delta x\right)^2\right] \le C\cdot O\left(\Delta x^2\right) \quad (19)$$

where C is a bounded constant related to the maximum value of $S''(x)$. In contrast to ReLU, whose second-order derivative is a pulse function leading to divergence and infinite variance in the loss function, GELU has an unbounded second-order derivative that amplifies variance in high-disturbance regions. However, the second-order derivatives of the cubic B-spline function are continuous and bounded, thereby avoiding the abrupt gradient changes observed in ReLU. Additionally, the derivative amplitude of the B-spline is regulated by the node spacing within each interval: $[t_i, t_{i+1}]$, providing greater stability compared to the higher-order derivatives of GELU. The local truncation error for the cubic B-spline is $O(\Delta x^2)$, reducing the overall error to $O(\Delta x)$. Since this analysis focuses only on B-spline curves and not surfaces, the error may increase further in multilayer neural network approximations due to potential accumulation.

To further improve the network's ability to capture robust features and mitigate the impact of adversarial examples, we introduce a MSDC, we can obtain a boundary-aware representation X_{MSDC} beyond the normal receptive field. The representation X_{MSDC} is then fed into the three-layer KAN, as follows:

$$y = \text{KAN}(X_{\text{MSDC}}) = (\Phi_3\circ\Phi_2\circ\Phi_1)X_{\text{MSDC}}. \quad (20)$$

Each KAN layer Φ consists of a univariate function matrix of $N_{\text{in}}\times N_{\text{out}}$, defined as

$$\Phi = \{\phi_{j,i}\} \quad i = 1,\ldots,N_{\text{in}}, \quad j = 1,\ldots,N_{\text{out}} \quad (21)$$

where each input X_i is transformed by a set of univariate functions $\phi_{j,i}$. The activation function that connects the ith node of the rth layer (r,i) and the jth node of the $(r+1)$th layer $(r+1,j)$ is given by

$$\phi_{r,j,i}\left(X_{r,i}\right) = w_b b\left(X_{r,i}\right) + w_s\text{spline}\left(X_{r,i}\right) \quad (22)$$

where $b(\cdot)$ is a basis function, spline$(\cdot)$ is a linear combination of B-splines, and w_b and w_s are the corresponding weights for the basis and spline functions. Here, we set

$$b\left(X_{r,i}\right) = \frac{X_{r,i}}{1+e^{-X_{r,i}}}. \quad (23)$$

Thus, the jth node value of the $(r+1)$th layer

$$X_{r+1,j} = \sum_{i=1}^{N_{\text{in}}}\phi_{r,j,i}\left(X_{r,i}\right) \quad j = 1,\ldots,N_{\text{out}}. \quad (24)$$

All nodes in layer $(r+1)$ are represented as

$$X_{r+1} = \begin{pmatrix} X_{r+1,1} \\ X_{r+1,2} \\ \vdots \\ X_{r+1,j}\end{pmatrix} = \begin{pmatrix} \phi_{r,1,1}\left(X_{r,1}\right)+\cdots+\phi_{r,1,i}\left(X_{r,i}\right) \\ \phi_{r,2,1}\left(X_{r,1}\right)+\cdots+\phi_{r,2,i}\left(X_{r,i}\right) \\ \vdots \\ \phi_{r,j,1}\left(X_{r,1}\right)+\cdots+\phi_{r,j,i}\left(X_{r,i}\right)\end{pmatrix}. \quad (25)$$

In summary, the function matrix Φ_r corresponding to the rth KAN layer is

$$\Phi_r = \begin{pmatrix} \phi_{r,1,1}(\cdot) & \phi_{r,1,2}(\cdot) & \cdots & \phi_{r,1,i}(\cdot) \\ \phi_{r,2,1}(\cdot) & \phi_{r,2,2}(\cdot) & \cdots & \phi_{r,2,i}(\cdot) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{r,j,1}(\cdot) & \phi_{r,j,2}(\cdot) & \cdots & \phi_{r,j,i}(\cdot)\end{pmatrix}. \quad (26)$$

We re-adjust the data dimension of the output features of the series of KAN layers so that their shape becomes $X\in\mathbb{R}^{B\times C\times H\times W}$, where B represents the batch size, C represents the number of channels, and H and W represent the height and width of the feature map, respectively. These features are subsequently processed by an efficient DwConv, which not only improves computational efficiency but also ensures the appropriate shape for the subsequent classification task. Formally, the output of the KAN module can be expressed as

$$X_{\text{output}} = \text{DwConv}\left(\text{KAN}\left(X_{\text{MSDC}}\right)\right). \quad (27)$$

Due to the smooth cubic spline-based activation functions within KAN, the network can capture intricate nonlinear interactions while maintaining smooth gradients. This property reduces the impact of adversarial attacks, making KAN particularly effective in adversarial settings. The combination of smoothness and expressive nonlinearity enables KAN to provide a robust solution for feature extraction, especially in tasks requiring resilience to adversarial perturbations.

IV. Experimental Results

The experiments in this article are organized into seven main sections. First, we provide a detailed description of the datasets used, highlighting their characteristics and relevance to HSI classification. This is followed by an explanation of the experimental setup, including implementation details, parameter settings, and evaluation metrics. Next, we evaluate the classification performance of our proposed method on three benchmark HSI datasets, comparing it with classical HSI classification approaches. We then investigate the impact of various attack methods and different perturbation magnitudes on classification performance. Moreover, ablation studies are conducted to dissect the contributions of individual components within our proposed method, offering insights into how each module influences overall performance and robustness. Additionally, we include experiments to assess the efficiency of feature extraction using KAN. Finally, we analyze the bright characteristics of first-layer class activation maps (CAMs) across different network architectures

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TABLE I
DATASET INFORMATION OF TRAINING AND TEST SAMPLES FOR EXPERIMENTS

Class	Salinas Valley				Pavia University				Indian Pines			
	Name	Training	Test	Color	Name	Training	Test	Color	Name	Training	Test	Color
1	Weeds_1	300	1709		Asphalt	300	6331		Alfalfa	20	26	
2	Weeds_2	300	3426		Meadows	300	18349		Corn-N	80	1348	
3	Fallow	300	1676		Gravel	300	1799		Corn-M	80	750	
4	Fallow_P	300	1094		Trees	300	2764		Corn	80	157	
5	Fallow_S	300	2378		Metal sheets	300	1045		Class-M	80	403	
6	Stubble	300	3659		Bare soil	300	4729		Class-T	80	650	
7	Celery	300	3279		Bitumen	300	1030		Class-P	15	13	
8	Grapes	300	10971		Bricks	300	3382		Hay-W	80	398	
9	Soil	300	5903		Shadows	300	647		Oats	10	10	
10	Corn	300	2978						Soybeans-N	80	892	
11	Lettuce_4wk	300	768						Soybeans-M	80	2375	
12	Lettuce_5wk	300	1627						Soybeans-C	80	513	
13	Lettuce_6wk	300	616						Wheat	80	125	
14	Lettuce_7wk	300	770						Woods	80	1215	
15	Vinyard_U	300	6968						Buildings	50	336	
16	Vinyard_T	300	1507						Stone	50	43	
Total		4800	49329			2700	40076			995	9254	

to evaluate and compare their robustness against adversarial attacks.

A. Data Description

In this study, three benchmark HSI datasets are employed to assess the performance of the proposed model. The detailed category information for three datasets is shown in Table I.

1) Salinas Valley: The first dataset was gathered by the airborne visible/infrared image spectrometer (AVIRIS) sensor over Salinas Valley (SV). The dataset consists of 204 spectral bands with a resolution of 512 by 217 pixels after excluding bands affected by interference. The data contains 16 crop categories, totaling 54 129 samples.

2) Pavia University: The second dataset was acquired by the reflective optics system imaging spectrometer (ROSIS) at Pavia University (PU) in Northern Italy in 2001, which consists of 103 spectral bands after excluding bands affected by interference with 610 by 340 pixels, and the data includes nine terrain classes, totaling 42 776 samples.

3) Indian Pines: The third dataset was collected by the AVIRIS sensor in North-Western Indian in 1992. After removing interference bands, it covers 200 spectral bands ranging from 400 to 2500 nm and has dimensions of 145 by 145 pixels. The dataset contains 16 vegetation classes, totaling 10 249 samples.

B. Experimental Setup

1) Evaluation Metrices: To assess the classification performance of each method, we utilized three evaluation metrics: overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ). All experiments were carried out with ten random repetitions, and the final classification results were obtained by calculating the mean and standard deviation across these repetitions.

2) Settings: The experiments described in this article were conducted using the Visual Studio Code platform on a single NVIDIA GeForce RTX 2080 GPU. The number of training and test sets was selected according to the configurations in Table I above, and the training process utilized the Adam optimizer with an initial learning rate of 5e-4, running for 1000 epochs.

C. Classification Performance Across Three Benchmark Datasets

In this section, we provide an in-depth analysis of the classification performance of the proposed BAN method and compare it with classical HSI classification techniques and advanced adversarial defense strategies. Specifically, we compare

BAN with spectral–spatial convolutional networks (SSFCNs) [39], Bi-LSTM [40], tokenization-based transformer frameworks such as SSFTT [41], vision transformer (ViT) [42], SpectralFormer [43], RSSAN [44], and specialized adversarial defense networks including SACNet [19], RCCA [20], and S^3ANet [22]. The robustness of our method is assessed by adversarial test samples on three widely recognized datasets, including SV, PU, and Indian Pines (IP).

Tables II–IV summarize the results of the classification comparison of our proposed method with nine other methods. Additionally, the results show that although many of the compared methods show significant accuracy degradation in the face of adversarial attacks, our BAN method maintains a high level of classification performance, and its classification results are very close to the real labels. This resilience underscores the robustness of BAN in not only achieving accurate classifications but also in maintaining stability when faced with adversarial perturbations.

Table II reports the quantitative results on the SV dataset. Traditional fully convolutional networks primarily focus on local feature extraction, rendering them more vulnerable to adversarial attacks. For instance, SSFCN and Bi-LSTM, which incorporate spatial and spectral information, have limited

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TABLE II

NUMERICAL CLASSIFICATION RESULTS FOR THE SV DATASET WITH THE BEST RESULTS HIGHLIGHTED IN **BOLD**

class	SSFCN	Bi-LSTM	SSFTT	ViT	SpectralFormer	RSSAN	SACNet	RCCA	S^{3}ANet	Ours
1(%)	**100.0±0.00**	63.55±12.48	44.37±19.65	38.66±7.30	49.26±8.22	39.71±12.66	99.99±0.02	99.99±0.02	**100.00±0.00**	**100.00±0.00**
2(%)	91.67±0.53	84.11±9.91	67.19±11.74	75.68±14.31	96.52±2.64	67.76±12.76	90.45±19.10	99.87±0.20	**99.95±0.04**	99.88±0.27
3(%)	3.97±0.58	46.29±16.44	51.28±18.29	40.48±6.11	51.78±14.92	49.44±14.13	92.54±13.74	99.70±0.27	**100.00±0.00**	99.99±0.04
4(%)	99.58±0.23	76.22±10.55	94.31±8.92	88.60±5.63	86.20±7.06	78.16±17.06	93.27±13.13	92.11±5.87	**99.71±0.26**	99.21±0.34
5(%)	31.01±18.12	84.68±12.47	44.28±12.38	72.96±15.23	60.06±10.54	76.55±11.69	83.01±30.03	97.16±1.24	98.83±1.99	**98.96±0.42**
6(%)	99.77±0.23	85.10±13.82	87.53±7.47	83.13±6.61	98.77±2.20	91.14±8.83	96.60±9.04	98.26±0.71	**100.00±0.00**	99.81±0.16
7(%)	3.43±0.39	70.27±10.34	81.51±12.22	51.31±8.28	95.78±1.15	65.94±17.92	96.80±7.39	98.00±0.93	**99.99±0.01**	99.96±0.03
8(%)	91.45±6.21	64.92±5.23	54.71±5.36	50.02±4.21	67.15±5.27	62.08±3.58	85.81±19.35	92.93±2.86	**99.28±0.82**	98.45±1.20
9(%)	8.71±5.23	81.56±19.91	72.46±12.59	65.89±18.38	84.49±7.35	59.36±22.30	91.49±21.26	99.53±0.09	83.56±11.16	**99.78±0.26**
10(%)	92.52±1.70	58.58±15.12	52.09±13.07	30.80±9.12	54.17±10.11	37.98±10.08	97.17±4.33	98.70±0.84	96.51±2.10	**99.90±0.11**
11(%)	87.63±6.67	27.22±9.85	19.71±12.65	25.99±5.93	23.67±16.22	22.49±15.20	99.80±0.18	99.16±0.58	**100.00±0.00**	99.67±0.19
12(%)	59.96±21.50	80.62±16.26	65.58±22.10	49.92±19.93	76.56±9.61	70.93±19.41	99.75±0.76	99.19±0.38	83.92±11.13	**99.86±0.10**
13(%)	99.82±0.05	81.19±13.55	22.58±10.37	8.96±10.71	59.04±19.04	27.07±10.91	**100.00±0.00**	98.50±0.54	**100.00±0.00**	99.97±0.10
14(%)	95.16±2.58	42.73±12.29	19.69±8.00	2.89±1.99	82.46±8.51	4.68±3.19	99.99±0.04	97.30±1.47	**100.00±0.00**	99.83±0.17
15(%)	10.23±2.61	58.97±9.79	46.98±6.45	46.88±9.20	54.43±11.59	59.15±11.55	58.23±45.20	22.92±8.82	**99.99±0.02**	99.74±0.28
16(%)	24.04±1.75	57.75±25.04	42.52±17.19	34.02±9.10	53.87±18.89	61.06±8.67	99.36±0.96	99.51±0.53	**100.00±0.00**	**100.00±0.00**
0A(%)	56.52±3.11	63.95±6.27	50.75±5.69	49.08±3.16	65.22±5.47	55.11±3.02	87.37±10.38	87.10±1.25	97.06±1.19	**99.48±0.26**
AA(%)	62.43±3.41	68.62±6.35	56.86±4.78	49.22±2.78	69.98±4.38	54.83±3.01	92.77±7.16	93.30±0.65	97.61±0.83	**99.69±0.10**
kappa(%)	51.01±3.48	60.26±6.78	46.58±5.76	43.97±3.39	61.67±5.85	50.47±3.23	85.79±11.64	85.51±1.41	96.70±1.33	**99.42±0.29**

TABLE III

NUMERICAL CLASSIFICATION RESULTS FOR THE PU DATASET WITH THE BEST RESULTS HIGHLIGHTED IN **BOLD**

class	SSFCN	Bi-LSTM	SSFTT	ViT	SpectralFormer	RSSAN	SACNet	RCCA	S^{3}ANet	Ours
1(%)	94.50±5.29	86.23±5.07	2.68±15.47	89.51±6.11	77.73±3.42	74.06±8.98	87.84±0.72	85.27±1.03	97.24±1.41	**97.28±0.39**
2(%)	28.65±1.47	91.43±1.73	83.77±2.88	90.53±1.70	89.53±3.55	90.80±6.74	87.51±4.22	91.10±1.19	94.48±2.56	**95.61±1.15**
3(%)	30.30±3.01	51.01±7.49	26.28±4.26	22.64±3.35	39.89±8.13	38.92±15.55	83.16±4.56	92.25±3.70	99.31±1.28	**99.81±0.20**
4(%)	**97.71±0.07**	74.08±12.02	75.41±5.91	75.32±9.46	63.66±17.66	59.98±11.21	91.51±0.89	70.40±1.60	96.46±0.34	96.03±1.15
5(%)	99.24±0.33	92.71±6.15	9.04±16.46	76.27±14.31	95.46±4.72	90.00±6.15	98.44±0.37	94.79±1.50	99.80±0.07	**99.98±0.06**
6(%)	68.38±0.78	37.52±2.66	9.13±13.29	52.75±4.92	50.89±8.39	36.74±4.24	90.82±2.20	97.41±1.45	**100.0±0.01**	99.37±1.27
7(%)	26.94±1.61	49.54±11.93	31.86±9.23	22.76±5.26	25.1±6.59	30.34±10.96	86.37±15.76	83.19±10.88	98.02±0.82	**99.68±0.51**
8(%)	81.69±1.35	78.46±7.74	54.42±10.29	44.80±13.19	70.12±3.59	48.30±12.68	90.40±1.61	93.62±1.22	94.95±1.67	**98.23±1.25**
9(%)	**100.00±0.00**	86.89±13.38	7.53±27.46	74.60±10.76	95.19±3.75	57.66±19.16	98.23±0.89	90.93±2.59	99.62±0.17	99.41±0.38
0A(%)	56.60±0.70	70.08±4.17	59.21±5.18	59.38±2.79	71.65±5.28	58.92±6.38	88.78±2.01	90.62±0.70	96.34±1.26	**97.03±0.87**
AA(%)	69.72±0.48	78.68±2.90	73.05±4.54	73.74±2.52	70.29±1.64	70.18±4.29	90.48±1.98	91.71±0.50	97.77±0.55	**98.37±0.62**
kappa(%)	48.57±0.73	62.80±4.80	50.60±5.47	51.36±2.96	63.74±5.23	50.55±7.25	85.34±2.50	87.68±0.90	95.18±1.63	**96.00±1.16**

TABLE IV

NUMERICAL CLASSIFICATION RESULTS FOR THE IP DATASET WITH THE BEST RESULTS HIGHLIGHTED IN **BOLD**

class	SSFCN	Bi-LSTM	SSFTT	ViT	SpectralFormer	RSSAN	SACNet	RCCA	S^{3}ANet	Ours
1(%)	93.75±0.00	0.00±0.00	12.65±5.10	18.85±6.41	7.78±8.79	5.80±4.48	**100.00±0.00**	91.15±5.72	98.75±3.75	**100.00±0.00**
2(%)	79.19±0.39	38.09±13.65	37.93±8.56	25.43±3.87	47.08±11.89	40.25±8.15	79.52±1.49	**95.24±0.98**	89.54±0.53	88.19±2.27
3(%)	80.03±0.51	12.43±11.44	16.45±2.82	16.19±3.54	41.07±9.66	27.08±10.98	82.90±2.00	93.13±2.58	**92.00±0.90**	89.45±1.37
4(%)	94.06±0.65	12.31±6.45	22.42±8.91	39.39±7.33	13.65±4.13	24.88±15.03	93.90±1.10	91.15±6.36	**96.74±0.44**	95.61±1.12
5(%)	90.62±0.94	45.93±24.87	46.32±14.17	35.09±7.18	33.59±10.62	32.49±12.07	96.05±0.80	93.70±3.50	**98.80±0.23**	94.99±1.84
6(%)	92.29±0.25	73.48±6.55	82.38±9.80	83.72±6.21	77.70±4.82	73.16±12.45	95.76±0.71	91.38±6.31	**97.65±0.29**	95.88±0.64
7(%)	**100.00±0.00**	0.00±0.00	7.69±15.13	50.75±22.24	7.84±9.81	7.44±11.23	**100.00±0.00**	90.00±10.91	**100.00±0.00**	**100.00±0.00**
8(%)	97.59±0.31	52.69±27.98	3.12±20.64	86.56±6.32	47.59±14.84	76.16±16.92	98.81±0.41	99.22±1.21	98.86±0.41	**99.91±0.21**
9(%)	**100.00±0.00**	0.00±0.00	7.00±12.61	43.43±26.65	9.43±7.21	10.08±14.46	**100.0±0.00**	**100.0±0.00**	**100.00±0.00**	**100.00±0.00**
10(%)	67.64±1.84	20.55±15.18	32.53±12.70	45.40±9.70	40.93±14.7	33.08±6.14	89.59±1.18	89.23±4.35	**94.69±1.13**	90.77±1.20
11(%)	69.59±1.25	37.56±19.81	62.48±5.95	57.75±5.62	68.38±12.21	50.57±7.35	72.51±7.03	84.61±4.65	90.64±2.31	**93.38±1.11**
12(%)	80.96±1.34	15.94±6.73	21.61±8.05	24.05±3.90	26.53±4.75	19.20±8.18	85.97±1.38	**89.94±2.38**	88.80±0.96	88.77±1.42
13(%)	98.71±0.00	44.59±14.37	8.54±16.35	53.99±20.73	45.34±12.87	40.22±16.59	**99.29±0.19**	98.80±1.61	98.71±0.58	98.65±1.10
14(%)	86.57±0.35	83.74±7.76	83.43±7.02	81.00±6.25	86.41±3.89	80.01±8.34	89.53±2.29	**97.07±1.86**	92.41±0.67	96.49±0.94
15(%)	81.01±0.46	40.37±24.46	1.07±15.39	61.98±9.87	45.50±9.40	34.74±9.31	99.14±0.49	98.96±0.78	**100.00±0.00**	99.64±0.42
16(%)	98.60±1.54	38.51±24.15	3.46±18.64	64.57±12.09	57.63±30.61	29.31±25.5	97.67±1.47	99.30±1.49	**100.00±0.00**	96.05±2.56
0A(%)	79.87±0.71	40.21±6.62	41.10±3.21	42.63±2.77	46.76±5.07	35.04±4.28	84.87±1.79	91.65±1.92	92.99±0.78	**93.13±0.59**
AA(%)	88.16±0.48	40.97±4.78	50.57±3.54	51.47±4.53	48.5±6.49	42.83±1.79	92.54±0.67	93.93±1.77	**96.10±0.54**	95.49±0.40
kappa(%)	77.08±0.80	33.19±6.58	34.82±3.21	35.18±2.98	40.34±5.25	28.91±3.52	82.82±1.99	90.40±2.20	91.98±0.88	**92.11±0.67**

performance in adversarial defense. Their classification accuracy drops remarkably when confronted with adversarial attacks. In contrast, transformer-based frameworks such as SSFTT, ViT, SpectralFormer, and RSSAN demonstrate varying levels of robustness. Notably, ViT achieves an OA of 49.08%, which is notably lower than some traditional methods like Bi-LSTM (63.95%). This suggests that transformers do not always outperform CNNs in terms of robustness against adversarial attacks. Otherwise, spectral–spatial self-attention and context-aware networks, such as SACNet, RCCA, and S^{3}ANet, exhibit better robustness against adversarial attacks. This indicates that focusing on global contextual information can mitigate the negative impact of adversarial samples. Our proposed BAN method shows significant improvements in OA, AA, and κ compared to the current state-of-the-art adversarial defense method S^{3}ANet across the three datasets. Specifically, on the SV dataset, BAN achieves enhancements of 2.42%, 2.08%, and 2.72% in OA, AA, and κ, respectively. These enhancements highlight BAN's ability to extract irregular regions with critical information in the integrated MSDC

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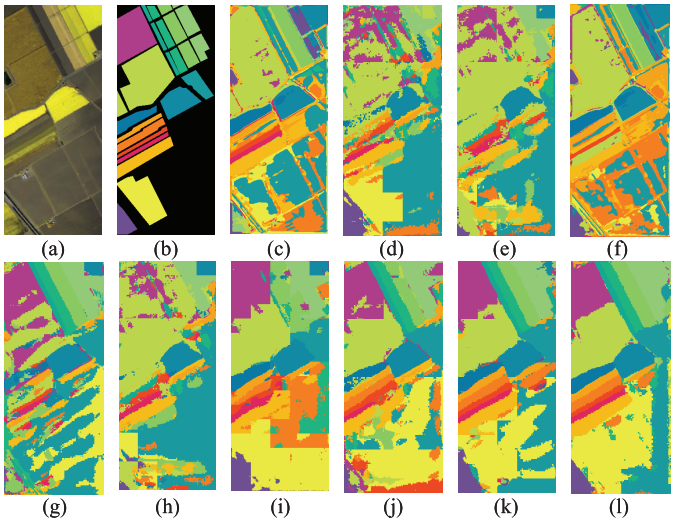


Fig. 5. Classification maps for SV dataset on the adversarial test set. (a) Original HSI image. (b) Ground truth. (c) SSFCN. (d) Bi-LSTM. (e) SSFTT. (f) ViT. (g) SpectralFormer. (h) RSSAN. (i) SACNet. (j) RCCA. (k) S^3ANet. (l) Ours.

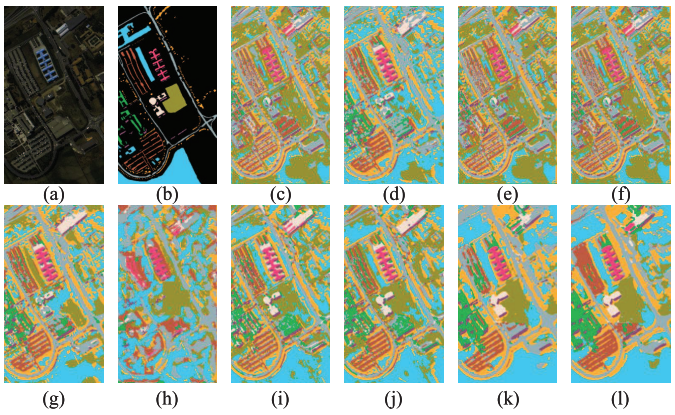


Fig. 6. Classification maps for PU dataset on the adversarial test set. (a) Original HSI image. (b) Ground truth. (c) SSFCN. (d) Bi-LSTM. (e) SSFTT. (f) ViT. (g) SpectralFormer. (h) RSSAN. (i) SACNet. (j) RCCA. (k) S^3ANet. (l) Ours.

and the advantage of utilizing KAN to maintain gradient smoothness. Similar conclusions can also be drawn from Tables III and IV of the other two datasets. For the SV and PU datasets, BAN achieves an OA exceeding 97%, and for the IP dataset, it exceeds 93% OA, representing the best results among all compared methods. Figs. 5–7 visualize the classification graphs generated by each of the ten methods. The majority of the methods show significant mispredictions under adversarial attacks, while the method proposed in this article achieves a classification performance closer to the true labels. Taking Fig. 5 as an example, according to the ground truth, the ninth-class region in the image belongs to the "Soil" category. However, most comparative methods misclassify this region as "Fallow" or "Lettuce," revealing a notable trend of misclassification. In contrast, the proposed BAN method generates results that are closer to the true labels and reflect the ground truth more accurately while suppressing edge perturbations.

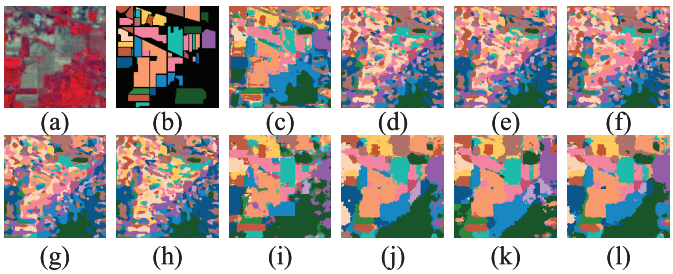


Fig. 7. Classification maps for IP dataset on the adversarial test set. (a) Original HSI image. (b) Ground truth. (c) SSFCN. (d) Bi-LSTM. (e) SSFTT. (f) ViT. (g) SpectralFormer. (h) RSSAN. (i) SACNet. (j) RCCA. (k) S^3ANet. (l) Ours.

D. Influence of Different Adversarial Attack Methods and Intensities

In this section, we use the FGSM adversarial attack method to generate adversarial samples with increasing attack strengths ranging from 0.01 to 0.1 to assess the impact of different attack intensities on the ten advanced HSI classification methods mentioned above. Fig. 8 shows the OA variation curves of these ten methods across three datasets as the strength of the adversarial attacks increases. The figure shows that as attack strength increases, the OA values of many methods decrease. This indicates that larger perturbations lower the network's accuracy on adversarial samples and weaken overall classification performance. By modifying the perturbation intensity, we can control the classification performance of the model, further reflecting the vulnerability of deep learning models in HSI classification.

To further evaluate the impact of adversarial attack methods other than FGSM on the ten advanced HSI classification networks, we also introduce two typical and efficient attack methods: PGD and C&W. For these three attack methods, we set the perturbation rate at 0.04, and both PGD and C&W are iterated ten times to generate adversarial samples. We also compared the classification performance on clean samples. Table V presents the OA classification accuracies under the three attack methods and one clean sample, where each group averages over ten runs with standard deviation as the result. We observe that under FGSM attack on SV, SACNet and S^3ANet achieve approximately 91.28% and 97.90%, respectively. However, when faced with PGD and C&W attacks, their performance sharply declines to below 60%. In contrast, our proposed BAN maintains over 90% accuracy under all three types of attacks. On the PU dataset, the method achieves the best OA for clean samples, as well as for FGSM and PGD attacks. It also ranks second among all methods on the IP dataset, with high OA for both PGD and C&W attacks. Compared to existing advanced methods, our proposed BAN shows enhanced robustness against both adversarial and clean samples.

E. Ablation Study

In this section, we evaluate the contribution of each module of the proposed BAN to the classification performance. The adversarial samples are generated using the FGSM attack, with the attack perturbation set to 0.04. To investigate the

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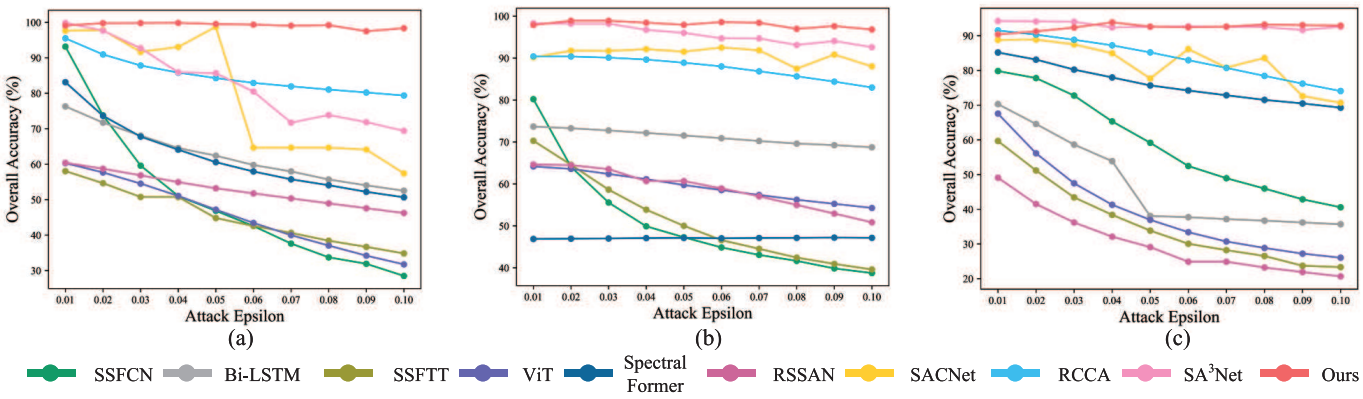


Fig. 8. OA of different methods on the adversarial test set with varying epsilon. (a) SV. (b) PU. (c) IP.

TABLE V

CLASSIFICATION OA OBTAINED BY DIFFERENT METHODS TESTED WITH CLEAN AND ADVERSARIAL SAMPLES IN THREE DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**

Method	SSFCN	Bi-LSTM	SSFTT	ViT	SpectralFormer	RSSAN	SACNet	RCCA	S^3ANet	Ours
Salinas Valley										
Clean Test Set	96.28±1.25	85.80±4.80	89.05±3.23	94.34±3.80	91.20±1.23	92.41±2.31	98.40±2.66	96.04±0.99	**99.98±0.00**	99.58±0.27
FGSM	53.23±1.51	63.95±6.27	49.41±4.10	49.08±3.16	67.26±3.56	55.11±3.02	91.28±4.98	87.86±1.46	97.90±1.18	**99.38±0.67**
PGD	10.96±1.97	40.89±8.58	36.97±6.00	33.77±2.86	39.58±4.28	41.20±4.39	39.74±26.07	88.01±1.43	43.14±2.66	**99.38±0.06**
C&W	20.69±3.22	44.08±7.50	34.78±6.03	26.78±1.91	41.75±3.64	39.92±5.30	52.39±20.96	83.75±2.44	51.56±3.69	**92.79±3.84**
Pavia University										
Clean Test Set	97.93±0.77	76.10±3.13	92.05±2.57	95.80±1.95	80.22±3.10	83.91±7.27	96.73±0.34	90.62±0.70	97.43±0.27	**98.79±0.45**
FGSM	51.86±1.52	70.08±4.17	56.76±4.26	59.38±2.79	68.89±5.30	58.92±6.38	81.86±6.63	89.61±0.58	96.70±0.26	**98.01±0.15**
PGD	25.23±0.80	56.79±3.19	48.89±4.89	52.54±2.29	58.00±6.41	41.24±6.44	74.88±3.55	89.28±1.68	31.28±1.59	**93.30±1.65**
C&W	26.80±0.43	58.62±2.86	49.65±4.68	51.71±1.55	58.53±4.59	45.64±5.68	76.36±4.87	**89.32±0.93**	31.98±1.69	84.55±7.79
Indian Pines										
Clean Test Set	82.17±0.43	46.41±6.60	82.01±5.05	91.81±3.06	60.15±1.78	63.95±13.51	87.75±0.50	**96.45±0.25**	92.46±0.30	92.52±0.16
FGSM	80.22±0.89	40.21±6.62	42.08±3.07	42.63±2.77	45.29±3.80	35.04±4.28	88.99±0.44	91.65± 1.92	92.39±0.14	**93.85±0.42**
PGD	22.98±1.31	32.35±5.60	28.80±3.55	29.20±1.18	17.86±3.72	20.47±8.71	52.64±4.77	**90.81± 0.27**	41.47±3.53	86.80±1.31
C&W	25.98±1.10	33.79±5.79	26.30±2.94	24.69±0.99	21.31±4.43	20.38±6.03	56.69±6.84	**91.71± 0.17**	41.43±1.42	87.85±0.49

TABLE VI

PERFORMANCE CONTRIBUTIONS OF DIFFERENT MODULES ON THREE DATASETS. THE VALUES IN PARENTHESES INDICATE THE GAP FROM THE BASELINE ACCURACY. **BOLD** NUMBERS REPRESENT THE BEST VALUES

Method	Salinas Valley			Pavia University			Indian Pines		
	OA(%)	AA(%)	Kappa(%)	OA(%)	AA(%)	Kappa(%)	OA(%)	AA(%)	Kappa(%)
Basebone	51.54	58.78	47.10	57.72	64.02	48.23	56.25	63.08	50.24
MSDC	89.58(↑38.04)	95.97(↑37.19)	88.33(↑41.23)	87.86(↑30.14)	89.93(↑25.91)	85.29(↑37.06)	86.76(↑30.51)	93.33(↑30.25)	85.94(↑35.70)
KAN	90.15(↑38.61)	90.44(↑31.66)	89.78(↑42.68)	98.01(↑40.29)	98.59(↑34.57)	97.31(↑49.08)	81.33(↑25.08)	86.30(↑23.22)	78.40(↑28.16)
BAN	**99.38(↑47.84)**	**99.36(↑40.58)**	**99.31(↑52.21)**	**98.28(↑40.56)**	**98.79(↑34.77)**	**97.68(↑49.45)**	**93.85(↑37.60)**	**95.85(↑32.77)**	**92.95(↑42.71)**

contributions of the MSDC and KAN modules to the classification performance of BAN, we designed ablation experiments on three datasets to validate the effectiveness of these two modules for HSI classification. Table VI presents the classification results of different modules, where the backbone represents the stages of the proposed framework. Specifically, we can conclude that both MSDC and KAN significantly enhance the robustness of the basebone in the presence of adversarial samples. Taking the SV dataset as an example, the OA without using the MSDC and KAN modules is only 51.54%. When MSDC is added, the OA increases to 89.58% (an improvement of 38.04%). Further incorporation of the KAN structure boosts the OA to 99.38% (an improvement of 9.80%). This indicates that the introduction of MSDC and KAN is crucial for enhancing adversarial robustness. Moreover, although MSDC achieves better performance on the IP dataset, its performance on the other two datasets is not as good as that of the KAN module. This suggests that while MSDC can significantly improve performance under certain conditions, it cannot fully replace the nonlinear modeling capability.

Fig. 9 presents the average spectral curves of the original HSI data and adversarial examples generated by different modules in BAN, used to analyze how each module affects the spectral reflectance characteristics and network robustness of adversarial samples. By comparing the spectral curves, it is evident that adversarial samples exhibit shifts in features compared to the original samples, particularly in certain bands of high-dimensional data where these shifts are more pronounced. Through the comparison of adversarial examples generated by different BAN modules, we can observe the effects of various defense strategies. Specifically, on the IP datasets, the average spectral curves of adversarial examples generated by the MSDC module closely match the original HSI data. Notably, the average spectral curves generated by the KAN module on the IP dataset are less similar to the original data

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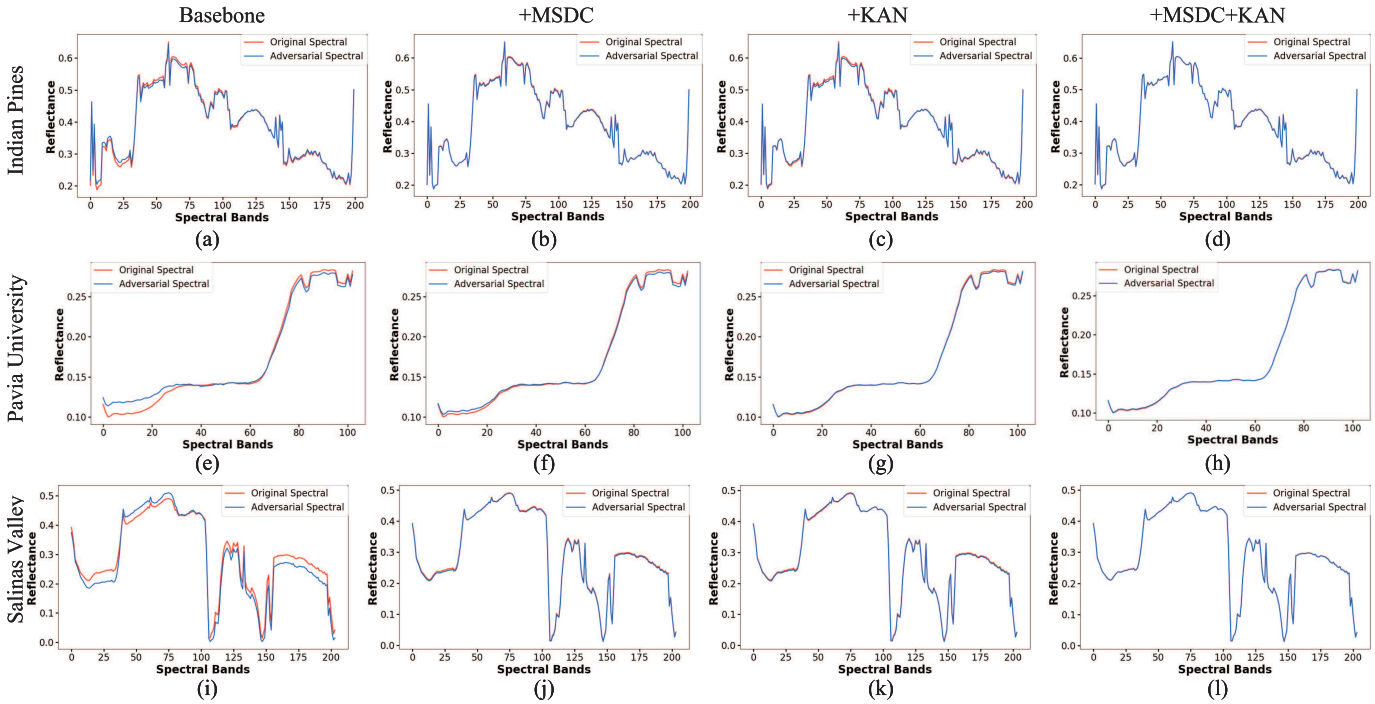


Fig. 9. Average spectral curve of the original HSI and adversarial examples generated under different modules in BAN on three datasets ($\epsilon = 0.1$). From top to bottom: (a)–(d) IP dataset, (e)–(h) PU dataset, and (i)–(l) SV dataset.

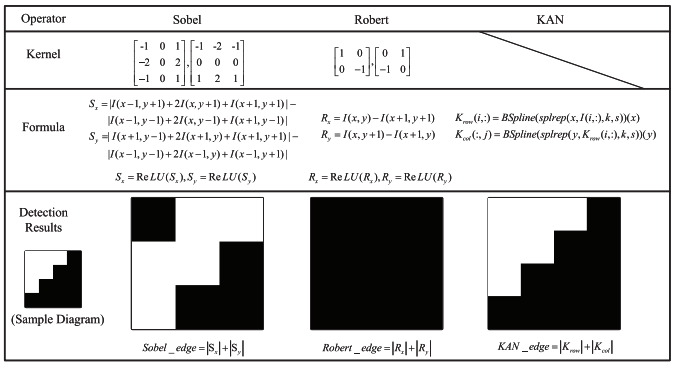


Fig. 10. Comparison of KAN and traditional convolution-based methods operators in edge detection performance.

compared to those generated by the MSDC module, consistent with the results in Table VI. Ultimately, only the adversarial examples generated by combining both the MSDC and KAN modules show highly consistent average spectral curves with the original HSI data across all three datasets.

F. Edge Detection Ability for KAN

In this section, we demonstrate the advantages of KAN in recognizing specific edge patterns within images through a simple edge detection example compared to traditional convolution-based edge detection methods such as the Sobel and Robert operators. As shown in Fig. 10, the purpose of this experiment is to detect oblique edges that represent significant brightness changes from the top-left to the bottom-right regions of an image. To simplify the experiment, we use a 2-pixel configuration of edges as a test benchmark and compare the performance of the three methods: Sobel, Robert, and KAN. As illustrated in the Fig. 10, both Sobel and Robert use convolution for edge detection, whereas the KAN algorithm is based on cubic spline functions to smooth the given image in both horizontal and vertical directions. The final edge detection result is obtained by calculating the sum of the absolute values of the smoothing results from both directions. Specifically, after performing convolution operations using the Sobel and Robert operators, we also incorporate ReLU nonlinear activation functions, respectively. KAN utilizes the splrep function to compute a B-spline representation of a given set of data points. This representation consists of nodes, coefficients, and the spline degree k, along with a smoothing factor s required to describe the B-spline curve. Subsequently, the B-splines at specific points x are interpolated using BSpline()(x) based on the B-spline representation obtained by splrep. Finally, edge detection is accomplished through thresholding. The results show that neither the Sobel operator nor the Robert operator can accurately detect specific edge regions. The Sobel operator does not perform accurately enough, while the Robert operator misclassifies in edge detection. In contrast, KAN's ability to accurately identify edge regions can be attributed to its inherent nonlinear smoothing and fitting capability, which enables it to detect edges in images more accurately than the linear models used in traditional convolution-based methods.

G. Comparative Analysis of CAMs

We compare the CAMs in the first stage to analyze the regions that different network architectures focus on when capturing feature maps in the first layer. By comparing clean samples with adversarial samples, we highlight the network's

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TABLE VII
COMPUTATIONAL EFFICIENCY COMPARISON OF TEN BENCHMARK METHODS ON THREE DATASETS

Method		Training Time (s)			Test Time (s)			FLOPs(M)			Params(M)		
		SV	PU	IP	SV	PU	IP	SV	PU	IP	SV	PU	IP
Patch-based	Bi-LSTM	45.63	16.10	3.83	19.97	23.70	28.64	42.70	17.81	41.57	1.64	0.52	1.58
	SSFTT	114.90	91.11	23.36	9.80	7.57	1.75	9.72	4.91	9.53	0.95	0.48	0.93
	ViT	75.96	35.33	42.18	37.30	61.11	6.12	47.61	45.41	50.83	0.97	0.93	1.04
	SpectralFormer	189.58	44.10	120.69	98.87	81.52	18.32	36.43	14.44	35.41	0.35	0.16	0.34
	RSSAN	77.04	40.47	54.77	37.12	68.52	7.03	4.75	3.31	4.70	0.11	0.07	0.11
Images-based	SSFCN	21.41	33.04	9.33	0.09	0.10	0.02	12157.83	19905.82	2295.61	0.11	0.10	0.11
	SACNet	26.60	56.97	13.66	0.10	0.12	0.03	18202.17	22016.05	3301.41	0.21	0.15	0.20
	RCCA	85.80	220.58	9.00	0.33	0.61	0.05	184109.71	421219.02	9147.53	1.22	1.42	0.90
	S^3ANet	234.94	266.13	222.17	0.36	0.40	0.29	21174.22	27807.66	3783.91	0.30	0.24	0.30
	Ours	122.73	241.05	35.61	0.24	0.40	0.07	4518.12	5017.12	838.83	0.18	0.17	0.18

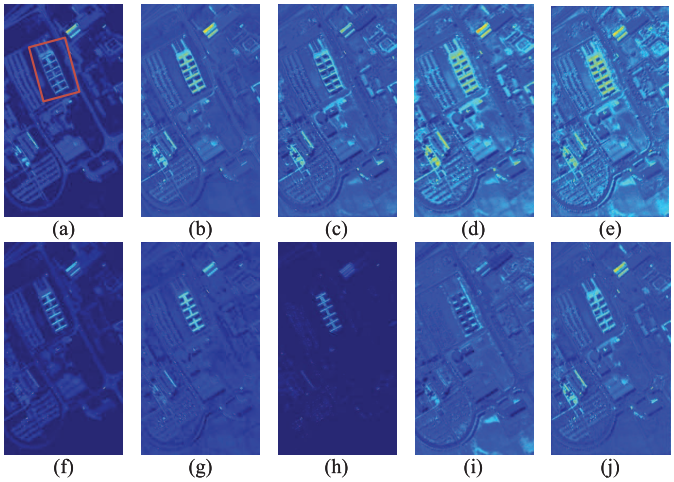


Fig. 11. Analysis of CAMs in the first layer and their sensitivity to adversarial perturbations. The area within the red rectangular box represents the feature map of the fifth class. (a)–(e) CAMs for clean examples, while (f)–(j) CAMs for adversarial examples. (a) and (f) RSSAN. (b) and (g) SACNet. (c) and (h) RCCA. (d) and (i) S^3ANet. (e) and (j) Ours.

sensitivity to robustness-related structures. Using the PU dataset, we compare the CAMs of the fifth class feature maps employing five different classification methods.

Fig. 11 presents the CAMs for clean feature maps generated by RSSAN, SACNet, RCCA, S^3ANet, and our method in the first row, and the CAMs for adversarial samples after adding perturbations in the second row. The results from (a)–(e) indicate that both S^3ANet and our method excel at capturing the most prominent focus areas in the brightest regions. However, after introducing adversarial perturbations, the results from (f)–(j) show a diminished capability in capturing key features, with the focus areas becoming smaller. Nonetheless, compared to other methods, our approach captures more significant focus areas in both clean and adversarial samples, demonstrating superior performance in capturing robust image features.

H. Comparative Analysis of Computational Efficiency

In this section, we present a comprehensive analysis of computational complexity across ten benchmark methods. As detailed in Table VII, we rigorously compare four key efficiency metrics: training time, testing time, floating-point operations (FLOPs), and parameters. The methods are categorized into two types based on the input type: patch-based methods (including Bi-LSTM, SSFTT, ViT, SpectralFormer, and RSSAN) and image-based methods (such as SSFCN, SACNet, RCCA, S^3ANet, and our method). It is worth noting that image-based methods typically have higher FLOPs than patch-based methods because they use the entire image as input. However, such methods have a natural advantage in modeling global contextual information, enabling them to more effectively capture long-range dependencies and spatial semantic relationships. Additionally, benefiting from modern hardware's efficient support for image-based operations, they exhibit stronger parallel computing capabilities, thereby demonstrating superior temporal efficiency in actual inference.

In image-based methods, our method achieved the lowest FLOPs across all datasets. Notably, S^3ANet also demonstrated robust adversarial robustness, but its training time on SV and IP was nearly twice that of our method, clearly demonstrating our significant advantage in training efficiency. Compared to patch-based methods, although they may exhibit some competitiveness in terms of training time and FLOPs, our method demonstrates exceptional balance and superiority in real-time capability and adversarial robustness, with testing time not exceeding 0.4 seconds. In summary, our method is not only more efficient in terms of computational resource utilization but also demonstrates greater adaptability and reliability in practical applications.

V. CONCLUSION

The vulnerability of DNNs to adversarial examples cannot be overlooked, especially in the boundary regions between classes. Existing defense methods often struggle to address these issues due to their inadequate adaptability to irregular boundary patterns and poor performance in handling large perturbations in boundary regions. To bridge this gap, we propose BAN, a boundary-aware network that combines the MSDC and KAN architecture. BAN utilizes multiscale deformable and dilated convolutions with learnable offsets to capture spatial–spectral features containing critical structural information at different scales. Moreover, by leveraging a KAN structure based on cubic spline functions, BAN reduces the impact of gradient-induced perturbations, thereby effectively alleviating adversarial perturbations in high-gradient boundary regions. The experimental results demonstrate that BAN achieves state-of-the-art performance under various types and intensities of attacks, highlighting its reliability in enhancing the adversarial robustness and security of HSI classification

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networks. However, our current defense mechanisms mainly focus on enhancing the robustness against gradient-based attacks, while attacks that do not rely on the gradient information of the model have not been fully explored and verified in our methods. Therefore, our future research will continue to explore this field.

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